Unit 2.1 Printables

DNA to Proteins (HS-LS1-1) · 8-day version: Siamese Cats

Everything students write goes in the notebook (Unit_2.1_Student_Notebook.pdf). These pages are only the extras.

Day	Print	Copies
3, 4, 7	Glue-in strips (page 2): Day 3 base key and code table, Day 4 code table, Day 7 Write It data	1 per student (cut apart)
3	Gummy DNA Lab directions (page 3) and mats (page 4)	1 directions page per student, 1 mat set per pair
4	Card sort, cards A to F (page 5)	1 set per group

Day 3 lab materials (per pair)

- 2 licorice strands (6 to 8 inches)
- 24 candy bases: 6 red (A), 6 yellow (T), 6 green (C), 6 orange (G), plus a few spare
- Toothpicks, a paper plate
- No food? Use colored beads and pipe cleaners, or colored paper circles

Cut on the dashed lines. Strips are sized to fit half of a notebook page.

Glue-in strips

Cut apart on the dashed lines and glue into the notebook.

Day 3 · glue in above Part 1: Lab record

Red = A Yellow = T Green = C Orange = G

DNA triplet	Amino acid	DNA triplet	Amino acid
ATG	Met	AGA	Arg
GGA	Gly	CTG	Leu
TTT	Phe	GAC	Asp

Day 4 · glue in at Part 2: Card sort

DNA triplet	Amino acid	DNA triplet	Amino acid
GGA	Gly	AGA	Arg
GAG	Glu	GTG	Val
GAA	Glu	CTG	Leu
CCG	Pro		

Day 7 · Part 3: Write It data, two Siamese kittens

	Kitten raised warm	Kitten raised cool
Ear skin temperature (°C)	34	26
Ear fur color	Light tan	Dark brown
Back skin temperature (°C)	36	35
Back fur color	Cream	Cream

Day 3 · Gummy DNA Lab: Directions

Question: How does a cell's copy of a gene become a chain of amino acids, and what happens if one letter changes? You will model the start of a cat's gene for a pigment-making protein. (The letters are an illustration, not the real gene.) Record your work in your notebook.

Materials (per pair)

- 2 licorice strands (the backbone)
- Candy bases: 6 each of red (A), yellow (T), green (C), orange (G), plus a few extra
- Toothpicks, a plate, the mats, the code table

Safety

- Do not eat lab materials until your teacher says you may.
- Toothpicks are sharp.
- Tell your teacher about any food allergy.

Part A: Build the template strand (6 min)

1. Attach bases to a licorice strand, left to right: **T A C C C T G A C A A A**
2. Write the colors you built.

Part B: Build the partner strand and the ladder (8 min)

1. On the second strand, pair each base (A with T, C with G).
2. Join the strands with toothpicks, twist gently, and have your teacher check it.
3. Which strand has the coding letters? How do you know?

Part C: Copy the message, then read it (10 min)

The cell **copies the gene's message** from the template strand, then the **copy is read** to link amino acids. The DNA stays safe.

1. Pull the strands apart. Keep the template.
2. Build a new strand by pairing with the template. This is the **copy**. Write its sequence.
3. Read the copy three letters at a time with the code table. Write the amino acid chain.
4. Explain the path in your own words, starting with the DNA.

Part D: Change one letter (7 min)

1. Swap the 4th base of the template (green C) for a yellow T. It now reads **T A C T C T G A C A A A**.
2. Build the new copy and read it. Write the amino acid chain.
3. Compare with Part C. How many amino acids changed? Which one?
4. Predict: could this protein work differently? Explain.

Connect to the Siamese cat

1. A one-letter change in the cat's gene swaps one amino acid. What did your one-letter change do to the chain?
2. Name two ways your gummy model is different from real DNA.

Clean up: trash the toothpicks and candy (or put them in the labeled container), wipe your table, and wash your hands.

Day 3 · Gummy DNA mats

Slide the licorice strand along the mat. One base per box, three boxes per amino acid.

Parts A and B: template strand (T A C C C T G A C A A A) and partner strand

Template

Partner

Part C: the copy

Amino acids: _____ – _____ – _____ – _____

Part D: changed template (T A C T C T G A C A A A) and its copy

Amino acids: _____ – _____ – _____ – _____

Day 4 · Card sort

Cards are 1.95 in wide to fit a notebook category column. Cut on the dashed lines, sort, and glue each card under its category on the Day 4 left page.

A In a Siamese cat, one letter changes in the gene for a pigment-making protein: GGA becomes AGA.	**B** A gene's triplet changes from GAG to GAA.	**C** A gene's triplet changes from CTG to CCG.
D One letter changes in a stretch of DNA that has no gene and no switch.	**E** One letter changes in a switch next to a gene so the gene can never be turned on, and its protein is not made.	**F** In a human gene, GAG becomes GTG.